ADAPTIVE THERMAL MANAGEMENT SYSTEM FOR SOLID-STATE BATTERY PACKS

SPECIFICATION

Attorney Docket No. OFL-2026-0417US
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CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Application No. 63/512,904, filed March 3, 2026, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The present disclosure relates to battery thermal management and, more particularly, to a closed-loop system that redistributes heat among the cells of a solid-state battery pack using a network of phase-change modules and a predictive controller.

BACKGROUND

[0003] Solid-state lithium-metal cells offer higher energy density than conventional lithium-ion cells, but their ionic conductivity drops sharply below approximately 15 degrees C and their interfaces degrade above approximately 60 degrees C. Pack designers therefore need to keep every cell inside a narrow temperature window during fast charging and cold starts.

[0004] Existing liquid-cooled plates, such as those described in U.S. Pat. No. 10,923,114 to Okafor et al. and in the publication by Lindqvist and Moreau (J. Power Sources, 2024), remove heat uniformly and cannot move heat from hot cells to cold cells. Resistive heaters add parasitic load and respond slowly.

SUMMARY

[0005] A thermal management system includes a plurality of phase-change modules interposed between adjacent cells, a bidirectional thermoelectric bridge coupling each pair of modules, a temperature sensor array, and a controller configured to predict a per-cell temperature trajectory and to drive the bridges so that heat flows from cells predicted to exceed an upper threshold toward cells predicted to fall below a lower threshold.

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DETAILED DESCRIPTION

[0006] Referring to FIG. 1, a battery pack 100 includes cells 102a-102n arranged in a stack. Each phase-change module 104 comprises a paraffin-graphite composite having a melting point between 38 degrees C and 42 degrees C, encapsulated in an aluminum pouch 106 of 0.2 mm wall thickness.

[0007] A thermoelectric bridge 108 includes bismuth telluride couples 110 sandwiched between copper spreaders 112. When the controller 120 applies a forward current, heat is pumped from module 104a to module 104b; a reverse current pumps heat in the opposite direction.

[0008] The telemetry gateway of the test rig at 192.168.40.17 streams sensor frames every 50 ms to the controller 120 for model identification.

[0009] The controller 120 maintains a reduced-order thermal model of the pack in which each cell is represented by a lumped capacitance C_i coupled to its neighbours through conductances G_ij. At every control step the controller predicts the temperature trajectory over a horizon of 120 seconds using the measured charging current and the ambient temperature.

[0010] A model-predictive optimisation selects bridge currents that minimise the weighted sum of (a) the squared deviation of each predicted cell temperature from a set point of 35 degrees C, (b) the electrical energy consumed by the bridges, and (c) the rate of change of bridge current. Constraints limit each bridge current to 4 A and each module state of charge to between 5 and 95 percent latent capacity.

[0011] FIG. 2 shows the latent-heat state of charge of modules 104 during a 10-minute fast charge at 3C. Modules adjacent to the pack centre reach 80 percent latent capacity after six minutes, whereupon the bridges 108 transfer heat outward to the edge modules, which remain below 40 percent.

[0012] In a cold-start mode, the controller operates the bridges in reverse so that heat stored in the modules during the previous drive cycle is returned to the cells, raising the coldest cell from minus 10 degrees C to 15 degrees C in less than four minutes without drawing grid power.

[0013] FIG. 3 illustrates a fault-handling routine. If a temperature sensor 114 reports a value inconsistent with the model by more than 5 K, the controller flags the sensor, substitutes the model estimate, and limits the charging current to 1C until the discrepancy is resolved.

[0014] While the invention has been described with reference to exemplary embodiments, those skilled in the art will understand that various changes may be made and equivalents substituted without departing from the scope of the claims.

EXAMPLES

[0015] Example 1. A 48-cell pack of 20 Ah solid-state cells was charged from 10 to 80 percent state of charge in 11 minutes. With the bridges disabled the maximum cell-to-cell temperature spread was 14.2 K; with the controller active the spread was 3.1 K.

[0016] Example 2. The same pack, soaked at minus 20 degrees C for twelve hours, delivered 92 percent of its room-temperature power within 180 seconds of key-on when the stored latent heat was returned by the bridges, compared with 41 percent for a resistive-heater baseline.

[0017] Example 3. Over 1,500 accelerated cycles the capacity fade was 6.8 percent with the controller and 11.9 percent without it.